

Anuja Tayal

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EDUCATION

University of Illinois, Chicago, United States
PhD in Computer Science

Aug 21 - Present

- **Advisor:** [Dr. Barbara Di Eugenio](#)
- **Relevant Courses:** Natural Language Processing, HealthCare/Biomedical NLP, Causal Inference, Graph Machine Learning

The LNM Institute of Information Technology, India
Bachelor in Computer Science

Aug 11 - May 15

RESEARCH INTERESTS

Conversational Systems, Reasoning, Knowledge Graphs, Neuro-Symbolic Systems, and Natural Language Processing for Social Good, particularly in Healthcare.

SELECTED PUBLICATIONS

Anuja Tayal, Barbara Di Eugenio, Thesis Proposal: A NeuroSymbolic Approach to Control Task-Oriented Dialog Systems In ACL'25 Student Research Workshop [\[Paper\]](#)

Anuja Tayal, Devika Salunke, Barbara Di Eugenio, Eulalia Puig Abril, Paula Allen-Meares, Carolyn Dickens, Olga Garcia-Bedoya, Andrew Boyd, Conversational Assistants to support Heart Failure Patients: comparing a Neurosymbolic Architecture with ChatGPT In LREC'26 [\[Paper\]](#)

Anuja Tayal, Devika Salunke, Barbara Di Eugenio, Eulalia Puig Abril, Paula Allen-Meares, Carolyn Dickens, Olga Garcia-Bedoya, Andrew Boyd, Towards conversational assistants for health applications: using ChatGPT to generate conversations about heart failure In Sigdial'25 [\[Paper\]](#)

Anuja Tayal, Barbara Di Eugenio, Devika Salunke, Andrew D. Boyd, Carolyn A. Dickens, Eulalia P. Abril, Olga Garcia-Bedoya and Paula G. Allen-Meares, A Neuro-Symbolic Approach to Monitoring Salt Content in Food In CL4Health workshop, LREC-COLING'24 [\[Paper\]](#)

Devika Salunke, **Anuja Tayal**, Barbara Di Eugenio, Paula G Allen-Meares, Carolyn A Dickens, Olga Garcia-Bedoya, Eulalia P Abril, Andrew D Boyd, Assessing Bias In ChatGPT's simulated clinical responses In Mitigating AI Risk through Ethical Data Science workshop, AMIA ([Accepted Oct 2023](#))

Anuja Tayal, Aman Tyagi, Dynamic Contexts for Generating Suggestion Questions in RAG Based Conversational Systems In PromptEng 2024 workshop, WWW [\[Paper\]](#)

Denisa Gandara, Hadis Anahideh, Matthew Ison & **Anuja Tayal**, Inside the Black Box: Detecting and Mitigating Algorithmic Bias across Racialized Groups in College Student-Success Prediction. In Education Research ([Submitted Nov 2022](#)) [\[Paper\]](#)

RESEARCH AND INDUSTRIAL EXPERIENCE

Research Assistant, UIC

May 22 - Present

Skills: LLMs, Neuro-symbolic systems, Personalization, Reliability, Tools, Dialog Systems, Agentic-AI, Augmentation, User-Study

Dialog Systems & LLM Architectures

- Demonstrated feasibility of a hybrid task-oriented dialog system + LLM architecture using a neuro-symbolic framework to handle numeric reasoning in health-related dialogs (monitoring salt intake).
- Achieved a +20% improvement in joint goal accuracy over LM-only baselines on a TODS benchmark, validating the advantage of neuro-symbolic control.

- Ran an IRB-approved user study ($N = 20$) with hospitalized heart failure patients, showing that neuro-symbolic and LLM-based dialog systems complement each other.
- Analyzing different persuasion techniques employed by the patient educator while providing education to heart failure patients.
- Demonstrating that integrating neuro-symbolic rules via dialog acts enables controllable dialog, where LLMs alone struggle to manage the dialog flow necessary in a healthcare setting.

LLM Synthetic Data Generation

- Examined different prompting strategies to generate personalised patient-oriented self-care conversations in low-resource settings.

Fairness, Bias & Responsible AI

- Modeled bachelor's degree attainment to analyze bias and fairness in educational datasets.
- Tested bias-mitigation techniques and identified persistent unfair outcomes across multiple fairness notions and demographic subgroups.

R&D Data Scientist Intern in NLP - PhD, P&G

Jun 23 - Aug 23

Skills: RAG, Few-shot, Prompting, LLMs, LangChain, LLM Evaluation, LLM Fine-tuning

- Addressed the problem of formulating suggestion questions in low-resource settings, by using retrieval augmented system
- Used Dynamic Shot Prompting to create Suggestion Question Generator

ML Engineer at Intain, India

Aug 19 - July 21

Skills: LM Fine-tuning, BERT, Similarity Matching

- Worked on extracting relevant information from diverse, unstructured documents.
- Designed and implemented automated deep learning language models to extract key attributes.

Researcher, Indian Institute of Technology (IIT), Indore

July 16 - Aug 18

- Optimized probe selection to find faults in networks that adapt to traffic dynamics.
- Used Graph ML and flow analysis to detect malware in botnet communication flows.

Software Developer, R&D Department, AMDOCS, Pune

June 15 - June 16

ACHIEVEMENTS

- Defended my Thesis Proposal **June 25**
- **Conferences:** WWW'24, LREC-COLING'24, SIGDial'25, ACL'25, GHC'25
- **Reviewer:** LREC-COLING'24, COLING'25, ARR July'24-Present
- **Program Committee:** PromptEng'26 (ACM WebConf'26 Workshop)
- **Creating AI-Enabled Health Team Data Fabric Hackathon** Team won 2nd place
 - Proposed Discharge Pathfinder, an LLM tool that streamlines NICU discharge by simplifying medical jargon to support caregivers and improve patient outcomes.
- CRA-WCP Grad Cohort for Women Travel Award'23, GHC Scholar'24, MSLD'25
- **Conference Volunteering:** EMNLP'22, EACL'23, NAACL'24, ACL'25
- **Teaching Assistant:** Mathematical Foundations of Computing, Languages and Automata, Program Design, Computer Organization and Architecture

REFERENCES

Dr. Barbara Di Eugenio Professor, UIC
 Dr. Natalie Parde Associate Professor, UIC
 Dr. Aman Tyagi Applied AI Researcher, Google